

Interactive Digital Art and Design SE608

Project Proposal Year IV

Student Name:	Caoimhe O'Brien	Student Number:	C00297178
Working Title:	Digital Accessibility Experience		
Description:	An interactive educational experience that highlights the different obstacles that various user bases face while navigating the online world. It will show why digital access equality is important and why the WCAG and similar guidelines are so important. This project will allow the user to play through different stories which highlight a specific set of issues for a player (such as visually impaired, non-native English speaker, mobility issues, connectivity issues). One of the main ways Accessibility is enforced is through WCAG 2 (Web Content Accessibility Guidelines). Within these guidelines there are 4 principles: Perceivable, Operable, Understandable and Robust. For my project, I will make at least 1 story/level for each principle. For example, for the "Operable" principle, I will make a game which highlights how it would be like to try to order something off a website that has poor navigation. This would highlight what it is like for people with low vision and also cognitive disabilities. Technical Elements: To create the experience, I will use Godot. It is a game engine which is optimised for both 2D and 3D and it allows you to code your game in C++ or C# rather than using its own scripting language (GDScript). Godot is also optimised on a multitude of devices such as mobile, Linux machines, desktop, etc. Godot allows for 2D skeletal animation (rigging) for the actual gameplay but the cutscenes will be animated in Blender which will then be exported into Godot. I will also be hosting this interactive experience online through itch.io and have a downloadable package through my portfolio (for windows, linux, mac, apk (apk) and .zip (for all others)). Art Elements: To create the assets for this game, I will use a mixture of Blender (for the environment/backgrounds, and Krita (for the 2D sprites). I also aim to create informational cutscenes using Krita/Blender to further inform the user on the importance of digital accessibility. Design Elements: For this app, I will be designing various menus and interfaces for the interactive experiences as well as creating a section on my portfolio website to allow people to play and download the experience onto their own machines.		
Reasons for selecting project:	This project will build on the work I did as part of my work placement in regard to accessibility. It is an important topic in the industry at the moment and this interactive experience aims to bridge the gap in understanding as a lot of the general public take accessibility for granted and/or are unaware of the impact of inaccessible web content.		

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Proposed research content:	https://dl.acm.org/doi/10.1145/3802826
External links (if applicable):	https://www.ohchr.org/en/instruments-mechanisms/instruments/convention-rights-persons-disabilities https://www.w3.org/TR/WCAG22/ https://www.iso.org/standard/51341.html https://www.iso.org/standard/71953.html https://nda.ie/publications/a-user-centred-exploration-of-the-assistive-potential-of-digital-voice-assistants-and-internet-of-things
Hardware requirements:	Lab computer, drawing tablet (already owned)
Software requirements:	Godot, Blender, C#, HTML/CSS/JS, GitHub, OneDrive, Krita, Itch.io,
Other requirements:	User testing (Ethics form)

Signed:

Date:

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Approved/Not Approved:	
Reasons for not approving project:	
Conditions attached to approving project:	
Approved/Not Approved:	
Name of Supervisor:	
Signed: _____	Date: _____